School of Computing and Artificial Intelligence

Faculty of Engineering and Technology

Sunway University

SOFTWARE REQUIREMENTS DOCUMENT

SEMESTER	: MAY 2025
COURSE NAME	: BIS2102 INFORMATION SYSTEM ANALYSIS AND DESIGN
LECTURER	: ASSOC. PROF. TS. DR. ASLINA BAHARUM
SYSTEM NAME	: Booking.com System
PROGRAM NAME	: Bachelor of Information Systems (Data Analytics)
CONTACT PERSON	: *LIANG YRENN DA* *22067128@imail.sunway.edu.my*

STUDENT ID	STUDENT NAME	GROUP	ROLE
22067128	**LIANG YRENN DA**	BIS 1-4	Project Leader
22071112	**YEOH JIE LONG**	BIS 1-4	System Analyst
22039069	**OOI JIN YON**	BIS 1-4	UI Coordinator
25018565	**WONG XU LIANG**	BIS 1-4	Tester / QA Lead
24082968	**MUHAMMAD HARIZ HAKIMI BIN MOHD HAFIZ**	BIS 1-4	UX Coordinator
22049043	**EMILY CHA YU YUE**	BIS 1-4	Backend Developer

DATE OF SUBMISSION	27/6/2025
ACTUAL DATE OF SUBMISSION	26/6/2025
NO OF DAYS LATE	0

Table of Content

1.0 Introduction 5

1.1 Background Study 5

1.1.1 What kind of system 5

1.1.2 Who needs the system 5

1.1.3 Why they need the system 7

1.1.4 How the proposed system can improve their activities 8

1.1.5 An existing or similar system developed for other organizations 9

1.2 Development Methodology 11

1.2.1 Requirement Analysis 11

1.2.2 System Design 11

1.2.3 Implementation 12

1.2.4 Integration and system testing 12

1.2.5 Deployment 12

1.2.6 Maintenance 12

1.3 Project Milestones and Deliverables 13

1.4 Project Schedules (Updated and Latest) 14

1.4.1 Task breakdown 14

1.4.2 Job assignments 15

1.4.3 Gantt Chart 17

1.4.4 Pert Chart 18

1.4.5 Critical path analysis 19

2.0 Functional Requirements 20

2.1 Overall UseCase Diagram 20

2.2 Details of Each Use Case 22

2.2.1 Use Case 1: AI Assistant Trip Planner 22

2.2.2 Use Case 2: Hotel Self-booking 26

2.2.3 Use Case 3: Transportation Booking 30

2.2.4 Use Case 4: Tour Guide Booking 34

3.0 Nonfunctional Requirements 37

4.0 Requirements Reviews 42

1.0 Introduction

1.1 Background Study

1.1.1 What kind of system

The aim of the Enhanced Booking.com is to be an AI-based travel planner and hotel reservation service, which will be the major extension of the current Booking.com mobile application. It is designed as a hybrid solution between transaction processing system (TPS), knowledge-based decision support system as well as a user productivity platform. It is based on the combination of artificial intelligence (AI) and machine learning, used to interpret the input received by the user, including the travel dates, budgets, preferences, and destinations, and create a real-time, personalized travel itinerary. This AI-suggested itinerary has lodging, travel means (ride-hailing app and buses), local attractions, and recommended activities or events. As opposed to traditional travel apps where users are required to do the search, comparison, and booking and even document management with multiple sources, the system offers an end-to-end planning and booking system through a single unified interface. It also has real time navigation capabilities in terms of GPS, integration of interactive mapping (e.g Google Maps), and up to date transport and event availability. The system is deployed as a cloud-based, mobile-first responsive design desktop application, supported by REST APIs to safely integrate with third-party applications (e.g. Klook integrations, payment gateway integrations, transportation database integrations). Finally, the system falls into the category of an all-comprehensive AI-driven travel management system that integrates automation, personalization, and booking features into one smart system.

1.1.2 Who needs the system

The suggested system is applicable to a wide range of users who may have different travel requirements and technological literacy. The most notable of these are individual leisure travellers who invariably find it difficult to use multiple sites or apps in order to organize trips. These users want convenience, affordability, and even reliability, which is encouraged by the personalized AI-generated itineraries the

system helps them to create. The process involves providing them with an organized and optimized, customized trip plan by filling in minimal information on trip preferences like the budget, travel dates interested activities preferences etc. The platform can also serve business travellers considerably. This is because , they have limited time and usually filled up with strict schedules and therefore need quick, effective systems to make flight, accommodation or transport reservations. With this, the AI helper could simplify the whole procedure and even propose a cost-effective or loyalty-related solution depending on the past booking records.

The system is also very useful to those who travel as a group such as families, friends, or even corporate delegations. Managing travel between multiple users may get expensive and tedious, and this system will reduce the pains by enabling all group members to view the itinerary, collaborate in making changes in real-time, and get notifications about each update. Besides, it makes sure the time of checking in/out of the hotel, group ticketing, and schedules of transport are coordinated and transparent to its participants. In addition, less experienced travellers or users of older generations, who might not be knowledgeable about digital interfaces, can enjoy a simplified and intuitive interface. The application allows the system to be within the reach of everyone by using onboarding wizards, in-app assistance, and visual navigation.

After that, direct users which uses the function of travel service providers, including hotels, car rental shops, bus agencies, and tour organizers, rely on the platform indirectly as well. When they become a part of the AI-powered ecosystem, these providers have the opportunity to reach a larger audience and enjoy the advantages of more intelligent user targeting via automated itinerary insertion. Tourism boards and local governments can also use the platform to market destinations, festivals, or eco-tourism projects, which once again increases the importance of the system throughout the tourism ecosystem.

1.1.3 Why they need the system

Firstly, even there are many travel planning apps on the market, travelers still encounter issues which the existing market does not address adequately. These comprise service fragmentation, data inconsistency, information overload and deficiency of contextual personalization. Usually, users need to go to various websites to use various services: there should be one site to book a hotel, another one to choose transport, another one to choose an event and one more to navigate a route. Not only it is time-consuming, but this type of a fragmented workflow also comes with risk (colliding schedules, inaccurate timelines of check-in, lost connections in transport). There have also been numerous complaints of unexpected fluctuation in price or availability, inadequate assistance during travel disruption, and instability regarding policies or local laws by many travellers.

These are the very pain points that have given rise to the necessity of having a single, smart platform. The enhanced Booking.com system empowers AI to provide a solution to them since when all travel activities are integrated into a single ecosystem, powered by contextual intelligence. The users are no longer obliged to look up the details of specific bus routes, make sure the hotel facilities are what they want, or schedule their attractions by hand. The system works with these variables and creates a customized solution eliminating the guesswork, as well as, human error. As an example, it could filter hotels that remain within the budget and are within a 10-minute walk of an appointment to access a museum with consideration of opening hours and weather condition. Such exactness minimizes the mental workload of the traveler significantly.

When real-time support is taken into consideration, the appealing nature of the value-added becomes further distinguished. In many cases, travelers are not able to predict uncertainties like flight delays, event cancellations, or closed roads. The system can review itineraries and re-book contingencies and keep the user informed without the need of manual entry as all of this can be done dynamically given AI capabilities and real-time data feeds. This timeliness flexibility is one of the more outstanding distinctions compared with traditional bookings systems. Moreover,

such functionality options as location-based offers, emergency SOS, and health or visa regulation compliance notifications offer not only convenience but security and assurance. In Short , these capabilities will answer a robust user demand to have an integrated, intelligent, and reliable travel assistance that cannot be fulfilled with conventional platforms.

1.1.4 How the proposed system can improve their activities

The system being proposed introduces a revolution in the interaction of a user with any travel planning through the automation of tedious routine activities, a more clarified understanding, and aided decision-making with the use of intelligent proposals. The user aspect of making the change is that it takes less time to plan and users have a more confident feel on the travel plan. A traveller would previously search through various hotels, visit sights in a new location, read reviews, and manually come up with a travel plan, which could take many hours. In this system, they only have to feed in their destinations of interest, amount of time they wish to stay and their interests and the entire program is presented with an estimated cost, methods of booking and sequencing that would save and utilize the most time possible. The journey may contain rational shifts between the hotel, attractions and restaurants, installed backup time on the trip, rest and feeding.

In addition, it would be easier to book accommodation, transport, and events in one workflow, thus boosting user efficiencies and minimizing transactions errors. Each booking updates live on the itinerary at large, instead of going in and out of apps. It is also possible to change the system in real-time, so when a traveller chooses to cancel a day trip or stay in a different hotel during a trip, the AI will be able to reapply its work to the new schedule and notify the service providers. The GPS and mapping APIs integration make it easier to navigate since it allows accessing the dynamic route directions, the expected time to travel and notification on the go. It is also making users prepared and safe because they may find emergency services or their embassies nearby when they decide to go somewhere and they have some urgent problem.

After that, enhanced also improves the functions of the system through its provision of predictive cost optimization. As an example, in case a hotel in one of the nearby districts will be a better value option, or in case the adjustment of travel dates would lower the prices on the air ticket drastically, the proactive change will be provided by the AI. This allows users to make more intelligent cost-effective decisions. Besides, the multi-language support, inline data validation, and in-app help chat contribute to a decrease in user errors and increase the ease of use, particularly in the less experienced travellers. Business users obtain expense claim and corporate records over the system, in the form of exportable summaries of trip details. Shared payment and collaborative itinerary editing are also group traveller-friendly features, making it possible to have synchronous and inclusive group travellers. All in all, the platform turns the process of travel planning into a streamlined, smart and personalized process.

1.1.5 An existing or similar system developed for other organizations

There are a number of already existing travel planning intelligent systems in the world market that have some features that would be similar to the proposed system. A good example is Wonderplan, an internet-based program that programs itineraries using the preferences of the user, span and place. Wonderplan uses the Google Places and other public APIs and provides simple hotel and attraction recommendations. Although it offers in-depth itineraries, it does not offer end-to-end integration. There are no in-app navigation or transport integration, and all bookings are carried out by third-party services. Wonderplan also does not have an adaptive dynamic itinerary management system and a real-time travel adjustment option which also confines it to be more of a recommendation tool rather than a dynamic travel companion.

There is one more example of a Layla, an artificial intelligence-based helper created by Roam Around. Layla also accesses via chat interfaces (Messenger or WhatsApp) and includes AI generative in answering travel questions and proposing trips. Whereas Layla is excellent in quick response time and interactive talking, it fails to generate systematic, editable itineraries, as well as booking integrations with the engines. It also does not have visual aids like maps or calendars, which may

make use in the case of multi-destination trips cumbersome. Layla can only be used as a chatbot and not a complete application, restricting its functionality and long-term viability used in elaborate travel planning.

Another AI tool in the travel market is GuideGeek, which provides location-based conversational experiences like travel advice, and employs real-time information to respond to location-related queries. It can be used when making small, situational travels (i.e. asking where the best coffee is nearby), but is not expected to be used to generate full-fledged itineraries or manage long-range travel. It also fails to reschedule, charge money or book. In contrast to our proposed system, GuideGeek does not support calendar displays, notification and user preferences being applied beyond individual queries.

Instead, the Booking.com-enhanced system is constructed as a complete, AI-native travel platform that has the best advantages of these systems and circumvents their shortcomings. It provides scheduled Itinerary creation, straightforward booking integration, live time s-rescheduling, and navigation. Its AI goes beyond answering questions, it allows predictive suggestions, flexibility in planning based on circumstances, and it learns upon interaction with the user. The platform also leverages the scalability and the credibility of the existing Booking.com infrastructure, such as possession of an enormous database of hotels, secure payment capabilities, and an experienced user channel. Moreover, the system has entirely graphic interface, map, calendar display, multi-language, and flexible dashboard that allows dealing with all the trip information in one place.

With these advantages combined, the system does not only compete but also outlines current solutions, which allows it to become a pioneer of next-generation travel technology. It closes the gap between booking and smart planning solutions, offering an end-to-end, seamless experience that is accurately focused on the needs of the real users.

1.2 Development Methodology

Requirements definition

- Users should be able to search for hotels, flights, or car rentals
- Bookings should be confirmed in real time
- The system should support user reviews, secure payments, and mobile access

1.2.1 Requirement Analysis

Functional:

- User login
- Search filters (location, price, rating)
- Booking process
- Payment

Non-functional:

- High availability
- Scalability to handle peak traffic
- Security

1.2.2 System Design

- Frontend design: User-friendly UI for desktop and mobile
- Backend architecture: REST APIs, database for hotels, bookings, users
- Technology stack: React.js (frontend), Node.js (backend), MySQL/PostgreSQL (database), AWS (hosting)
- Third-party integration: Payment processors, map services, SMS/email notifications

1.2.3 Implementation

- Modules are built for login, search engine, booking, and payment.
- APIs are developed to fetch available rooms, prices, and process bookings.
- Admin dashboard for hotels to list rooms, manage pricing and availability.

1.2.4 Integration and system testing

- Unit testing: Each module (search filter, payment) tested in isolation
- Integration testing: Ensure data flows correctly between modules.
- System testing: Test full workflow — from user search to booking confirmation.
- Security and performance testing

1.2.5 Deployment

The system is launched in a live environment:

- Backend and database deployed to cloud (AWS or Azure)
- Domain and SSL set up
- CI/CD pipeline may be used for seamless deployment
- Beta version may be tested by internal users before full launch

1.2.6 Maintenance

Ongoing support and updates:

- Fixing bugs (booking errors, payment failures)
- Updating features based on user feedback
- Ensuring system uptime and performance
- Applying security patches and keeping integrations up to date

1.3 Project Milestones and Deliverables

No.	Date/Week	Milestones	Deliverables
1.	4	Submission of proposal	Proposal
2.	7	Submission of SRS	SRS
3.	8	Submission of SDD	SDS
4.	10	Submission of STD	STR
5.	12	Submission of Final Report	Final Report
6.	12	Presentation	Presentation

1.4 Project Schedules (Updated and Latest)

1.4.1 Task breakdown

No.	Tasks	Duration
	Project Initiation	**2 weeks**
1.	Requirements Gathering	1 week
2.	Create Proposal	1 week
3.	Business Case	1 week
4.	System Request	1 week
5.	Proposal mini presentation	1 week
	SRD Writing	**2 weeks**
5.	Gantt Chart & CPM	1 week
6.	SDLC	1 week
7.	Fact Findings	1 week
8.	SRD mini presentation	1 week
.	**SDS Writing**	**2 weeks**
9.	DFD Diagram	2 weeks
10.	Data Dictionary	2 weeks
11.	Use Case Diagram	2 weeks
12.	Class Diagram	2 weeks
13.	Sequence Diagram	2 weeks
14.	State Transition Diagram	2 weeks
15.	Activity Diagram	2 weeks

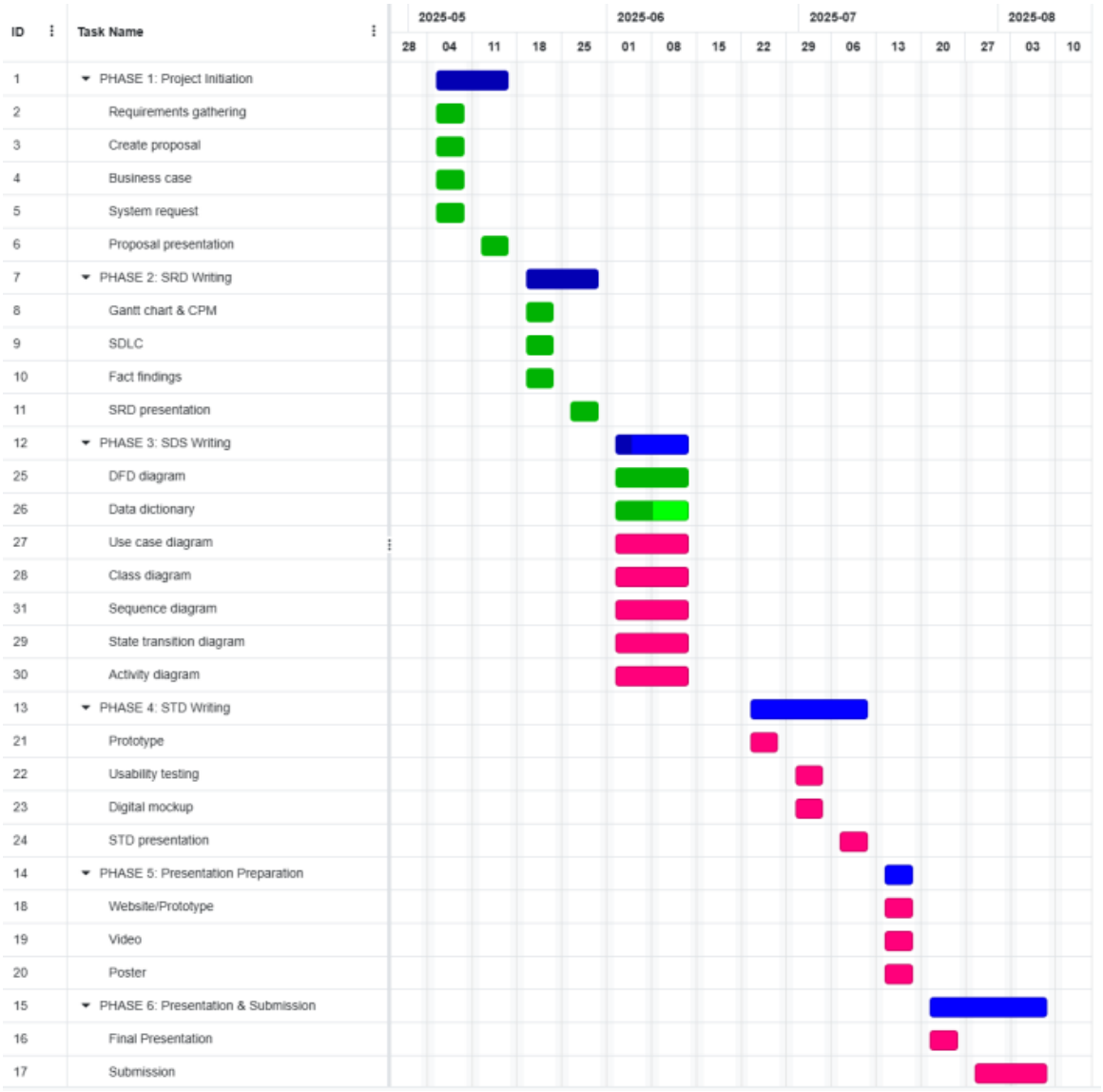
	STD Writing	**3 weeks**
16.	Prototype	1 week
17.	Usability Testing	1 week
18.	Digital Mockup	1 week
19.	STD mini presentation	1 week
	Presentation Preparation	**1 week**
20.	Website/Prototype	1 week
21.	Video	1 week
22.	Poster	1 week
	Presentation & Submission	3 weeks
23.	Presentation	1 week
26.	Submission	2 weeks

1.4.2 Job assignments

No.	Name	Job	Description
1.	Liang Yrenn Da	Project Leader/Developer	• Lead team • Assigning job to team members • Develop system
2.	Yeoh Jie Long	System Analyst	• Analyzing from Data Gathering • Support design planning
3.	Ooi Jin Yon	UI Coordinator	• Create and

			manage Figma designs • Lead structure and component design
4.	Hariz Hakimi	UX Coordinator	• Plan user journey and interactions • Support Figma prototyping •
5.	Emily Cha Yu Yue	Backend Developer	• Design database • Handle backend logics
6.	Wong Xu Liang	Tester / QA Lead	• Test system functionality • Log and report bugs

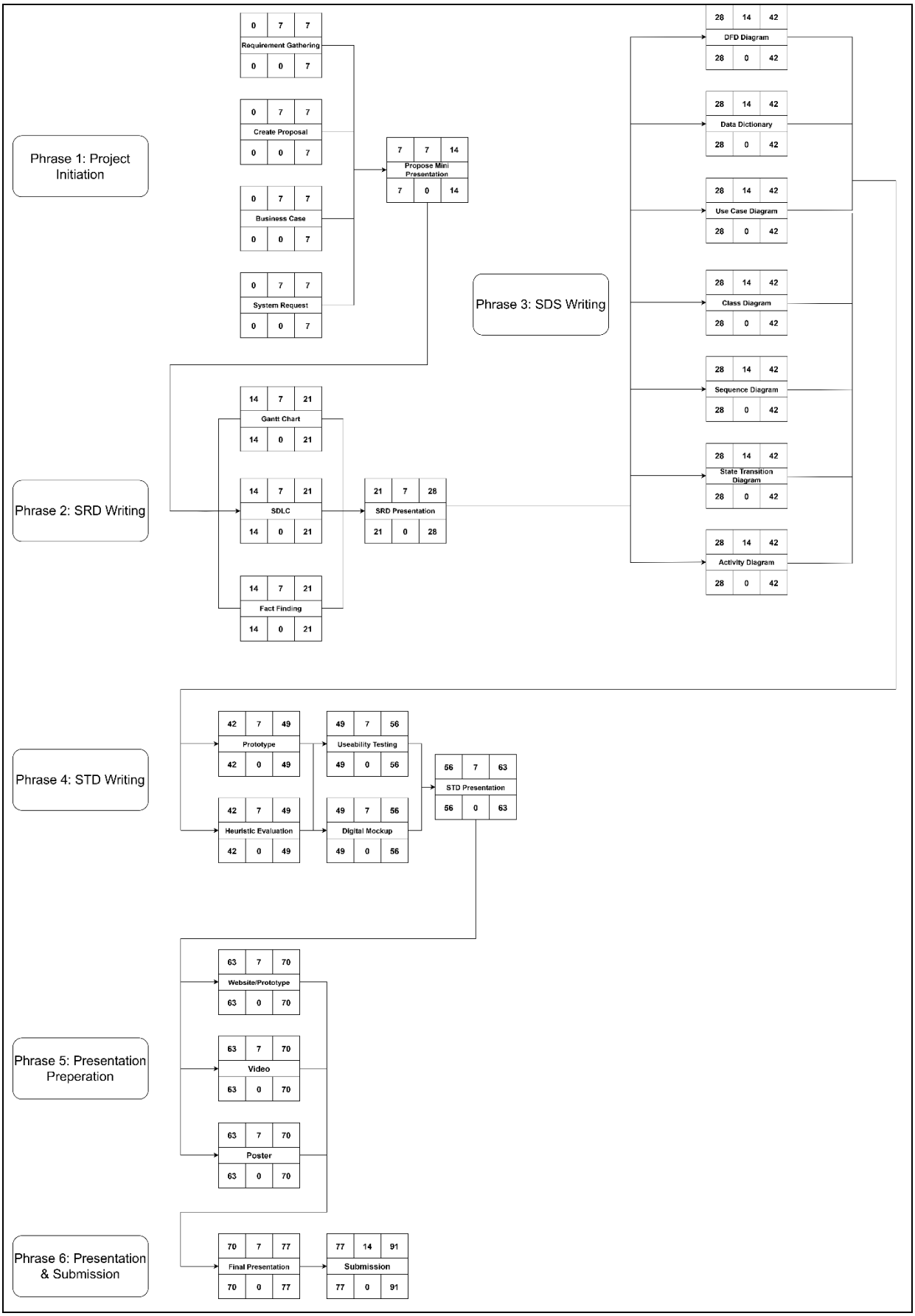
1.4.3 Gantt Chart



1.4.4 Pert Chart

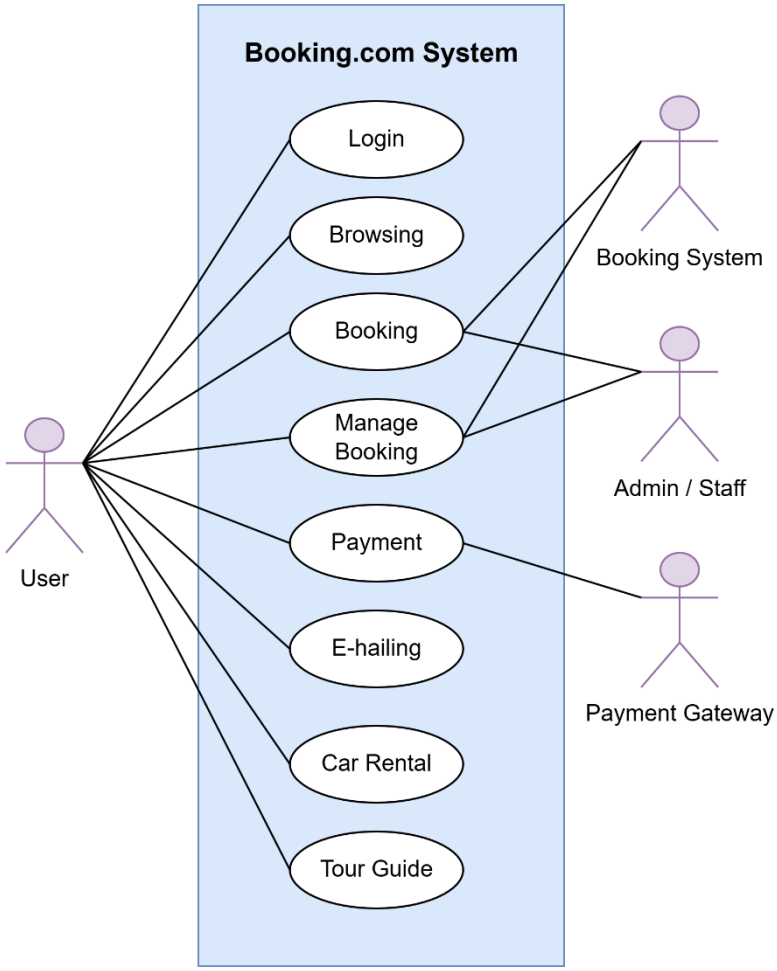
TASK	Predecessor	Optimistic (O)	Most Likely (M)	Pessimistic (P)	Expected Duration (TE)
Requirement gathering		4	5	7	5.17
Create proposal		4	5	7	5.17
Business Case		4	5	7	5.17
System Request		4	5	7	5.17
Proposal presentation		4	5	7	5.17
Gantt chart		4	5	7	5.17
SDLC		4	5	7	5.17
Fact finding		4	5	7	5.17
SRD Presentation		4	5	7	5.17
DFD Diagram		9	10	12	10.17
Data Dictionary		9	10	12	10.17
Use Case Diagram		9	10	12	10.17
Class Diagram		9	10	12	10.17
Sequence Diagram		9	10	12	10.17
State Transition Diagram		9	10	12	10.17
Activity Diagram		9	10	12	10.17
Prototype		4	5	7	5.17
Usability Testing		4	5	7	5.17
Digital Mockup		4	5	7	5.17
STD mini presentation		4	5	7	5.17
Website/Prototype		4	5	7	5.17
Video		4	5	7	5.17
Poster		4	5	7	5.17
Final presentation		4	5	7	5.17
Submission		9	10	12	10.17

1.4.5 Critical path analysis



2.0 Functional Requirements

2.1 Overall UseCase Diagram



Users

Users are the customers of Booking.com. They must log in first before accessing the application. After that, they can start browsing for hotels, flights and others. They can make a booking and make their payment. After that, they can manage their booking. In addition, they can also use the e-hailing function to call the taxi or use car rental function to rent their car in their destination. Lastly, they can book for a tour guide to take them travel across the city or country.

Booking System

The booking system is used to confirm the user’s booking and help to process any changes made by the user.

Admin or Staff

The admin will examine the booking, and any changes made by the airline or hotel companies.

Payment Gateway

Payment gateway is used to confirm whether the payment made from customer is valid or not.

2.2 Details of Each Use Case

2.2.1 Use Case 1: AI Assistant Trip Planner

Brief Description

A user can use an AI Planner to get a suggested itinerary by entering budget, destination and duration.

Precondition

Before this use case can be initiated, the User has the account to access the Booking.com application.

Postcondition

At the end of the use case, the User will be given a satisfying itinerary and let them book their trip.

Normal Flow

Steps	User	System
1	The user opens the Booking.com application	
2	The user logs in to the application	
3		The system displays Log In successful. (Ex1)
4	The user selects AI Trip Planner function.	
5	The user will enter	

	the budget, destination and duration.	
6		The system will provide the suggested itinerary. (Ex2)
7	The user will confirm their itinerary	
8	User will enter their personal details to ensure the booking is associated with related person	
9	User will look through their booking overview	
10	The user will proceed to the payment	
11		The system will confirm the booking. (Ex3)
12	The user booking is done.	

Exception/Alternative Flow

Ex1

The system will display log in unsuccessfully.

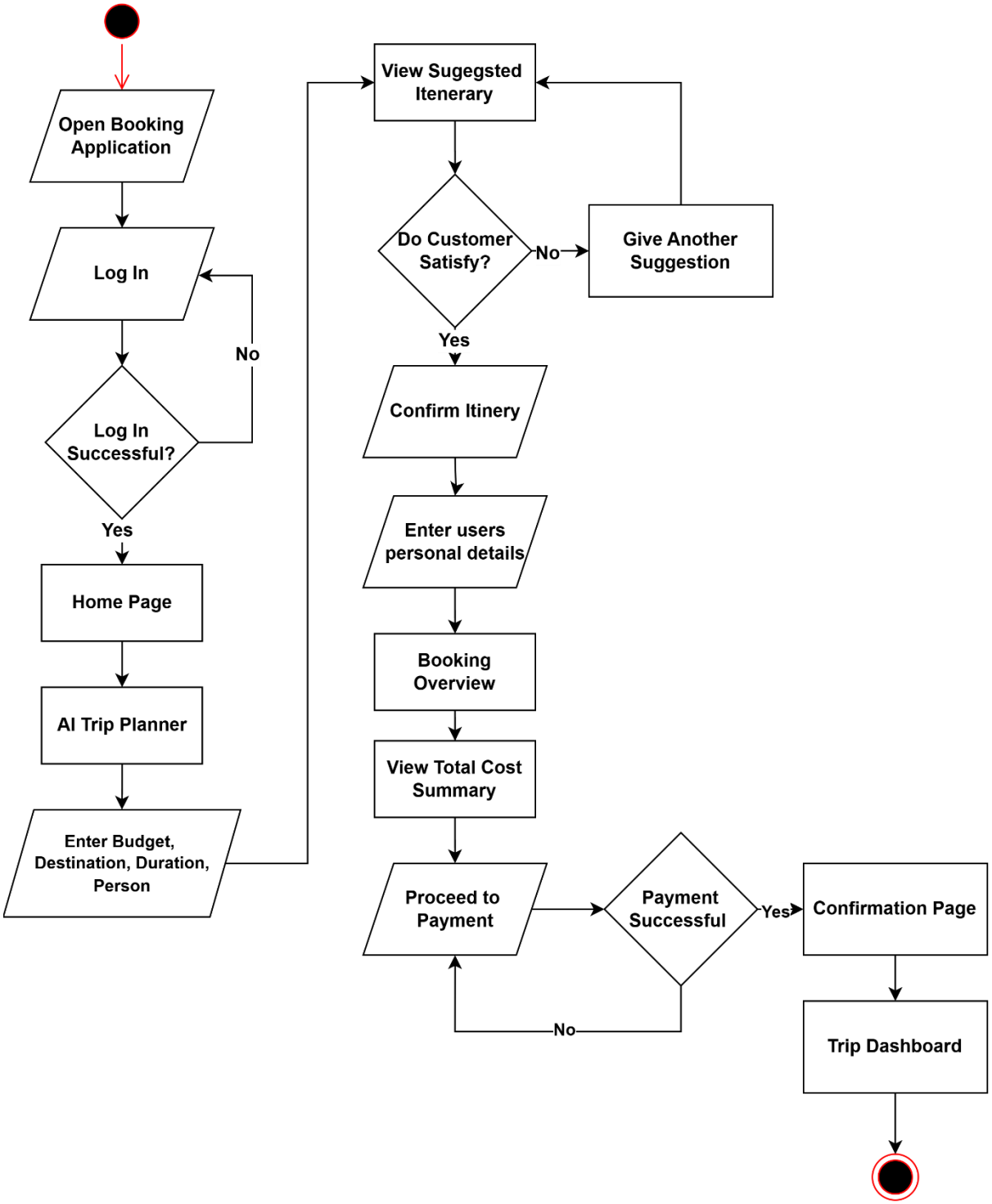
Ex2

The AI assistant will provide a new itinerary if the customer is unsatisfied.

Ex3

If the transaction is unsuccessful, it will reject the booking.

Activity Diagram



2.2.2 Use Case 2: Hotel Self-booking

Brief Description

A user can also search their hotel by searching for the location, date and time, number of people and filter function. They can find the best matched hotel they wanted to book.

Precondition

Before this use case can be initiated, the User has the account to access the Booking.com application.

Postcondition

At the end of the use case, the User will find their favorite hotel and book it for their trip.

Normal Flow

Steps	User	System
1	The user opens the Booking.com application	
2	The user logs in to the application	
3		The system displays Log In successful. (Ex1)
4	The user manual finds their hotel.	
5	The user will enter the number of persons, duration and their destination.	

6		The system will provide the hotel listing.
7	The user will click onto the hotel page (Ex2)	
8	The user booked the hotel	
9	The user will confirm their itinerary	
10	User will enter their personal details to ensure the booking is associated with related person	
11	User will look through their booking overview	
12	User will proceed to the payment	
13		The system will proceed with the payment. (Ex3)
14		The booking is successful.

Exception/Alternative Flow

Ex1

The system will display log in unsuccessfully.

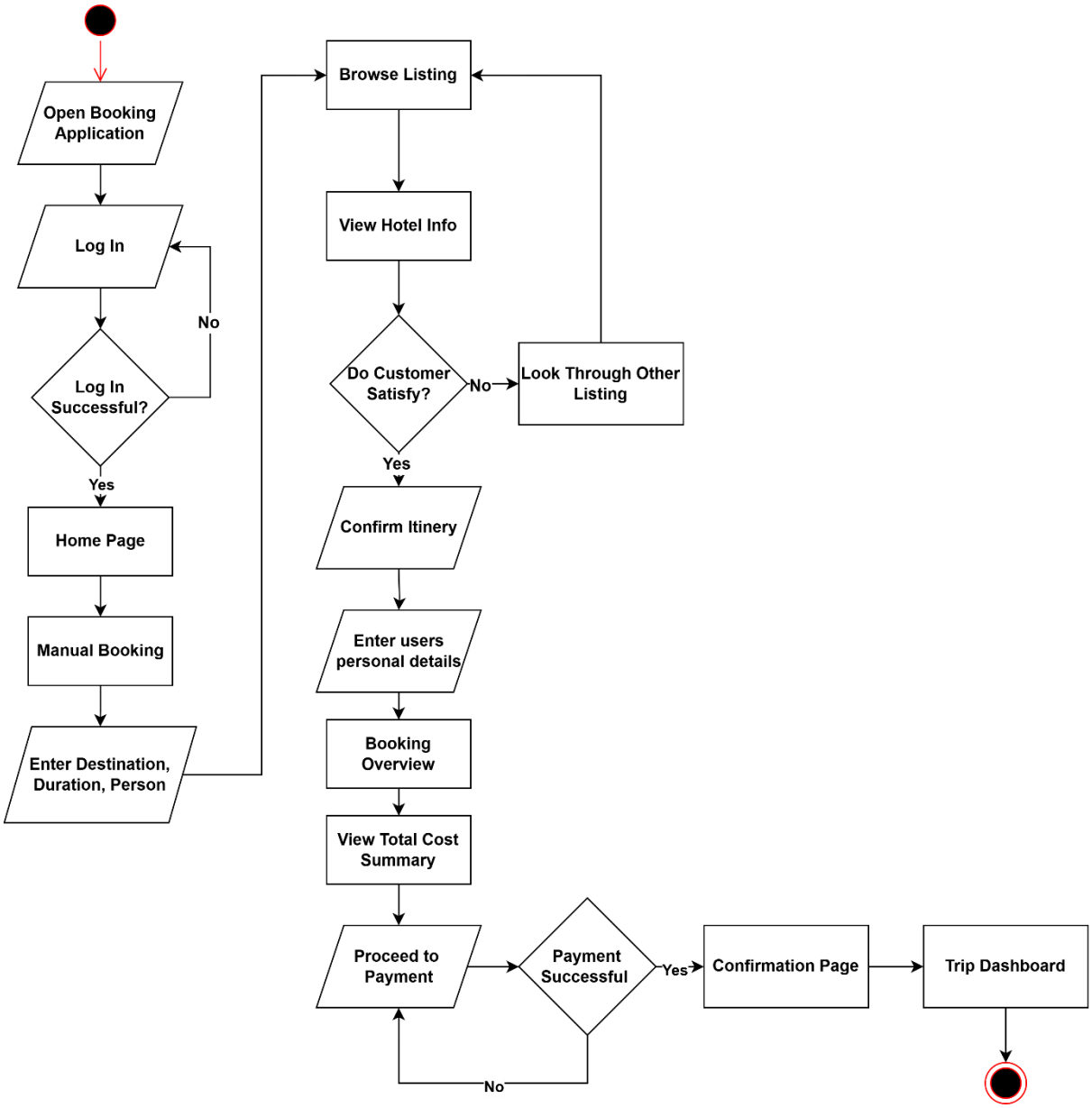
EX2

If users are not satisfied with the current hotel page, they can return back to room listing to search for another hotel.

Ex3

If the transaction is unsuccessful, it will reject the booking page.

Activity Diagram



2.2.3 Use Case 3: Transportation Booking

Brief Description

A user can select transportation such as car rental, bus or taxi. The car rental is only available for some countries and able to let them choose the car model. Besides that, they can also buy the local bus service for different companies to travel across cities. Finally, they can use the taxi service to go to another destination within the city.

Precondition

Before this use case can be initiated, the User has the account to access the Booking.com application.

Postcondition

At the end of the use case, the User will find suitable transportation to take them to their destination.

Normal Flow

Steps	User	System
1	The user opens the Booking.com application	
2	The user logs in to the application	
3		The system displays Log In successful. (Ex1)
4	Users can choose between renting a car, buy the bus ticket or call for pick-up via taxi.	

5	**Car Rental:** Users select a suitable car and rent it **Bus:** Users find a suitable bus by entering their destination and number of people. **Taxi:** They enter the pick-up and drop-off points and enter the number of people.	
6		The system will provide a list of options for users to choose based on the transportation users choose.
7	Users pick one of the options and confirm the itinerary (Ex2)	
8	User will enter their personal details to ensure the booking is associated with related person	
9	User will look through their booking overview	
10	Users bring to total cost summary and payment section.	

11	Users will proceed with the payment	
12		The system will proceed with the payment. (Ex3)
13		The booking is successful.

Exception/Alternative Flow

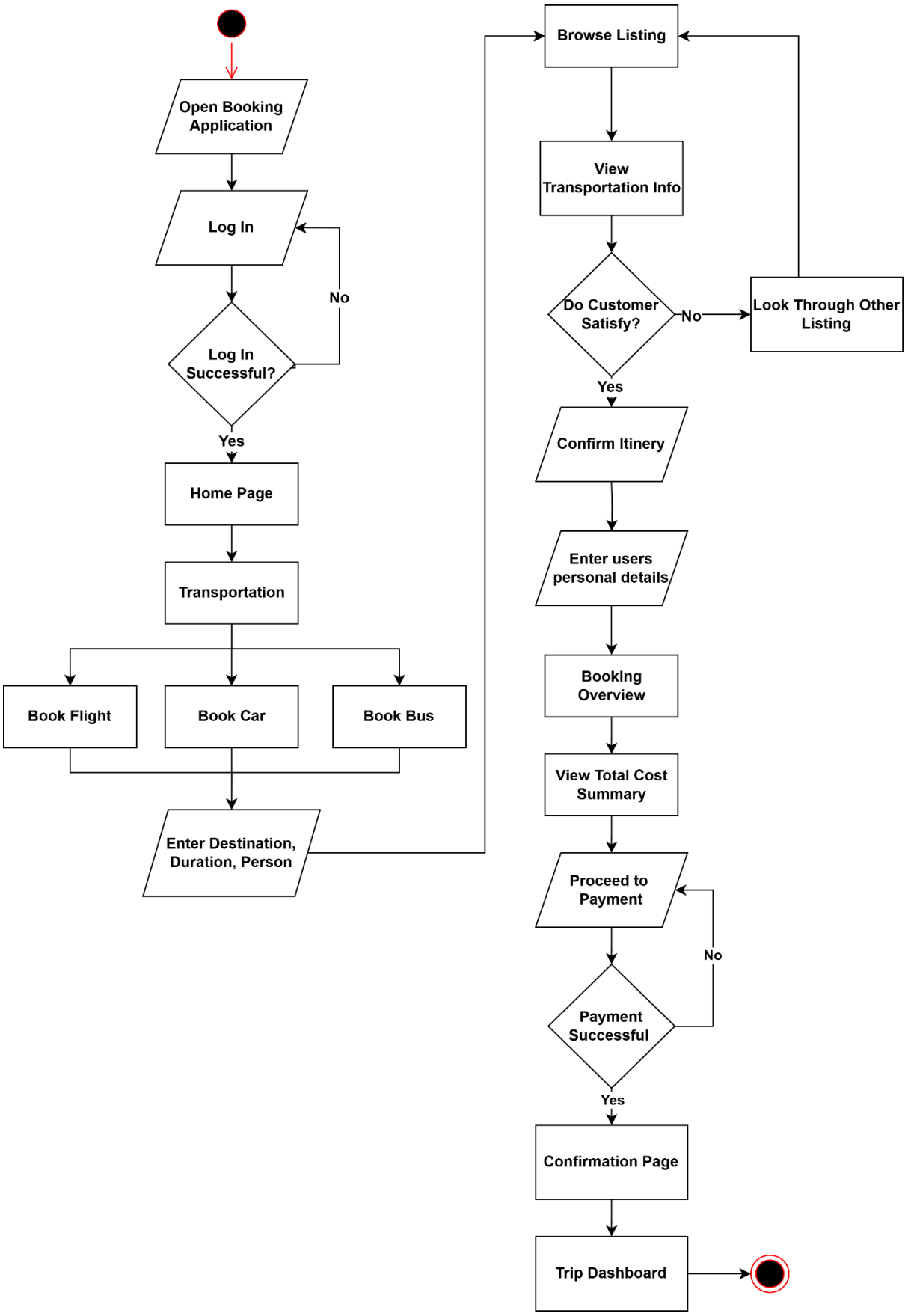
Ex1

The system will display log in unsuccessfully.

Ex2

If the transaction is unsuccessful, it will reject the booking page.

Activity Diagram



2.2.4 Use Case 4: Tour Guide Booking

Brief Description

The users are able to book a tour guide to travel across the city or country. They able to choose their preferred languages.

Precondition

Before this use case can be initiated, the User has the account to access the Booking.com application.

Postcondition

At the end of the use case, the user will find the tour guide to help them travel across the country and know better about a country.

Normal Flow

Steps	User	System
1	The user opens the Booking.com application	
2	The user logs in to the application	
3		The system displays Log In successful. (Ex1)
4	Users select tour guide booking system	
5	Users enter the destination and duration	
	Users enter their preferred language	

	and the type of tour	
6		The system will provide a list of options for users to choose variable of tour guide information
7	Users pick one of the tour guide from information page (Ex2)	
8	Users confirm their itinerary	
9	User will enter their personal details to ensure the booking is associated with related person	
10	User will look through their booking overview	
9	Users bring to total cost summary and payment section.	
10		The system will proceed with the payment. (Ex3)
11		The booking is successful.

Exception/Alternative Flow

Ex1

The system will display log in unsuccessfully.

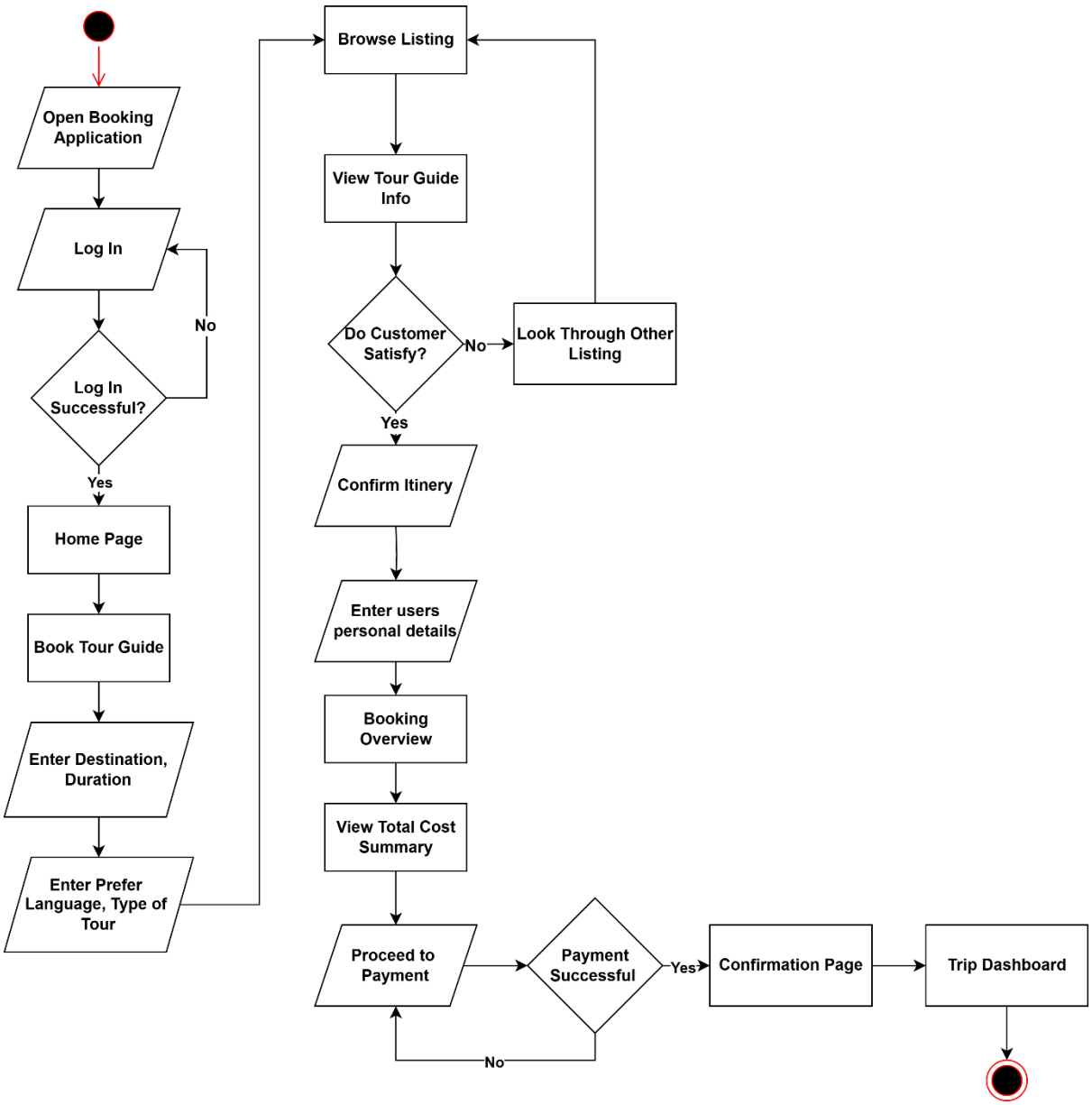
EX2

If users are not satisfied with the current tour guide page, they can return back to room listing to search for another hotel.

Ex3

If the transaction is unsuccessful, it will reject the booking page.

Activity Diagram



3.0 Nonfunctional Requirements

Nonfunctional Requirements	Description
Performance	1. **Generation Latency**: Under typical conditions, the system should generate a complete draft itinerary (including hotels, transportation alternatives, cost estimate, and timetable) within 3 seconds of receiving user input. 2. Justification: According to most User behavior studies/research, it tells us that the response time of under 3 seconds is acceptable for task that require background computation or data aggregation. If its over 3 seconds, users tend to lose focus or assume the system is really slow. 3. “3 seconds is about the limit for users to feel like the system is responding quickly enough to keep their attention focused.” (Source: Nielsen Norman Group – Response Times: The 3 Important Limits) 4. **UI responsiveness**: All user-initiated operations (filtering options, opening detail panels, map launches) must be completed within 200 milliseconds. 5. Justification: When a system is slow to respond (1000ms and above), it will degrade usability of the user and lead to the user being frustrated or irritated. Therefore, a proper response time of 100 – 200 ms is necessary. Google's RAIL model states that in order for a user interaction to feel instantaneous, it should be reacted to in 100–200 milliseconds (Google, n.d.).

	6. **Throughput**: The backend planning API must handle at least 5,000 concurrent requests without compromising response-time SLA (for example, 3 seconds max). 7. Justification: This is in line with load requirements for platforms with strong demand, particularly during peak hours.
Reliability and Availability	1. **Uptime Target:** More than 99.7% availability per month (< 2 hours downtime) 2. Justification: While most platforms target 99.9%, we feel that 99.7% is still strong and gives more time for maintenance/enhancement. Not only that, but this also ensures high availability that minimizes revenue loss and preserves user trust. 3. **Graceful Degradation**: The system will continue to create and show the remaining itinerary if an external service (like hotel, bus, or map APIs) is unavailable, making it obvious to the user which parts were not able to be retrieved because of the service outage. For instance, display the itinerary without the hotel information and a notice such as "*** Hotel suggestions are temporarily unavailable. Try again later" if hotel data is not accessible. 4. Justification: Users' trust is preserved when the system partially fails rather than completely breaking down. It also prevents user frustration if the system completely crashes, especially important in travel planning.

	5. **Disaster Recovery Planning**: Schedule regular backups of vital data, such as user profiles and itineraries, to ensure secure storage solutions. This approach ensures that data may be recovered in the event of catastrophic failures. 6. Justification: In the event of unplanned system failures or outages, backups guarantee that critical company operations (such as user itinerary access, trip planning, and customer assistance) can restart with the least amount of disturbance. Revenue and user confidence are safeguarded in this way.
Compatibility	1. **Client Support:** 2. **Web**: latest major versions of Chrome, Firefox, Edge and Safari 3. **Mobile**: responsive design for widths 320–1920 px or hybrid app on iOS 13+ and Android 9+.
Capacity	1. **Storage Needs**: 2. Initial footprint: ~500 KB per user (profiles + itineraries), ~50 MB for service catalogs. 3. Plan for 10× growth in 2 years → ≥ 100 GB total. 4. Justification: Planning for capacity growth is crucial to avoid performance issues, downtime, or data loss as user base expands.
Security	1. **ISO/IEC 27001:2022 certification:** establishes a formal, verified Information Security Management System (ISMS) for people, processes, and technology, which

	Booking.com and other global platforms rely on. 2. Justification: A certification indicates that your company has recognized, evaluated, and is actively addressing information security threats. Booking.com uses this specific certification across all of its functionality on its platform. Therefore, having this certificate for the new system will ensure data security and privacy for both users and Booking.com. 3. **SOC 2 Type II Compliance:** Showcases the continuous operational effectiveness of your security, availability, and confidentiality controls—essential for gaining enterprise and partner trust. 4. Justification: vital for SaaS platforms managing connectors and consumer data, particularly when catering to large clients. 5. **PDPA (Malaysia) Adherence Adheres**: local legal standards, including breach notification timescales, data subject rights, and retention restrictions, ensuring compliance with Malaysia's Personal Data Protection Act and regional roll-out. 6. Justification: Compliance guarantees that the platform satisfies local legal standards for data protection and user rights.
Usability	1. **Multi-Language Support**: The system should support at least three languages (e.g., English, Bahasa Malaysia, and Simplified Chinese), with the ability to switch between them at any time.

	2. **Learnability**: First-time users will be taken via a contextual onboarding wizard that takes less than two minutes. 3. **Error Prevention**: Inline validation for unsupported country selections (e.g., car rental coverage) and missing hotel check-in/out times, with contextual explanation and a "request time" button. 4. **Accessibilit**y: Meets WCAG 2.1 AA standards, including keyboard navigation, screen-reader labelling, and color contrast ratios of 4.5:1.
Maintainability & Manageability	1. Modular Design: Core functions (itinerary creation, booking adapters, and user management) must be separated into microservices with well-defined APIs. 2. Justification: Coupling is decreased by distinct service boundaries. Without having to closely coordinate deployments with other teams, teams working on user administration, booking integrations, or itinerary development can fully own their services. 3. Observability: Centralised logging (ELK/EFK stack) and distributed tracing (OpenTelemetry) are in place; health dashboards will show critical metrics (error rates, latency, request volume). 4. Justification: Logs and traces aid in promptly determining which service failed, where it occurred, and why problems arise (such as unsuccessful itinerary generation or a faulty API call). It also minimises Mean Time to Detect (MTTD) and Mean Time to Resolve

	(MTTR) and avoids protracted debugging sessions.

4.0 Requirements Reviews

Use Case Name	Completed by	Reviewed by	Review Results
AI Trip Planner (2.2.1)	Yeoh Jie Long	Wong Xu Liang	Completed
Hotel Self-Booking (2.2.2)	Wong Xu Liang	Emily Cha Yu Yue	Completed
Transportation Booking (2.2.3)	Emily Cha Yu Yue	Yeoh Jie Long	Completed
Tour Guide Booking (2.2.4)	Wong Xu Liang	Emily Cha Yu Yue	Completed